HYDROTECHGROW: SMART HYDROPONICS

PROJECT REPORT PHASE II

20ITPJ801

Submitted by

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BONAFIDE CERTIFICATE

Certified that this Project Phase II report **HYDROTECHGROW: SMART HYDROPONICS** is the Bonafide work of **BHARATH L (412522205018), GOKULAKANNAN V (412522205041), RAKESH V (412522205105)** who carried out the **20ITPJ801 – PROJECT PHASE II work** under my supervision.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ACKNOWLEDGEMENT

A successful man is one who can lay a firm foundation with the bricks others have thrown at him.
— *David Brinkley*

Such a successful personality is our beloved founder Chairman, **Thiru. MJF. Ln. LEO MUTHU.** At first, we express our sincere gratitude to our beloved Chairman through prayers, who in the form of a guiding star has spread his wings of external support with immortal blessings.

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We thank all the teaching and non-teaching staff members of the Department of Information Technology and all others who contributed directly or indirectly for the successful completion of the project.

ABSTRACT

The goal of our project is to automate the whole process of hydroponic cultivation using various sensors and cloud service platforms to monitor and control them.

Hydroponic systems have emerged as a sustainable and efficient method for crop cultivation, offering controlled environments and optimal nutrient delivery. In this project, we present a novel approach to elevate hydroponic farming through the integration of advanced sensor technologies and computer vision. Our system incorporates sensors to monitor critical environmental parameters such as temperature, humidity, Total Dissolved Solids (TDS), and pH levels in real time. These sensors provide continuous data streams enabling precise control and maintenance of optimal growth conditions for various plant species.

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CHAPTER 1

INTRODUCTION

Farmers face increasing challenges in managing their land and ensuring crop productivity during natural and man-made calamities, such as floods, droughts, and erratic weather patterns. These challenges are exacerbated by climate change, depleting natural resources, and population growth, all of which threaten global food security.

Traditional farming methods, heavily reliant on soil quality and stable environmental conditions, are proving insufficient to meet the growing demands for sustainable and efficient agricultural practices.

We believe that hydroponics represents the future of agriculture, offering a promising solution to these pressing challenges. Hydroponics eliminates the dependency on soil by growing plants in a nutrient-rich water solution, significantly reducing water wastage and the risks posed by unfavourable soil conditions. As an advanced farming method, it addresses the dual issues of limited arable land and the need to maximize crop yields within confined spaces, making it particularly suitable for urban and semi-urban environments.

Our approach integrates hydroponics with the principles of Controlled Environment Agriculture (CEA), leveraging IoT and cloud-based management systems to automate and optimize the farming process. Real-time monitoring of parameters such as temperature, humidity, pH, and nutrient levels ensures optimal growing conditions for plants. Cloud connectivity allows farmers to access data and control systems remotely, enabling efficient farm management even during adverse conditions.

1.1 PROBLEM STATEMENT

One of the biggest issues facing the agricultural sector in India is low yield: India's farm yield is 30-50% lower than that of developed nations", as stated by STARTUPINDIA. There are many reasons for this low productivity in crops. Agriculture continues to depend on monsoons, an unpredictable source of water for crops.

Another reason for low productivity is the size of farms in India. There is also a lack of timely information on topics such as beneficial crop sowing patterns and market price trends that would help farmers make the right decisions.

Traditional farming methods face several challenges, including excessive water usage, dependency on soil quality, susceptibility to pests and diseases, and the labour-intensive nature of crop maintenance. With the increasing demand for sustainable agricultural practices, there is a need for a smart and efficient solution that can optimize resource utilization, monitor plant health, and ensure consistent crop yield, especially in areas with limited arable land or water resources.

Hydroponics, a soil-less cultivation technique, offers a promising alternative; however, existing systems lack automation, real-time monitoring, and disease management capabilities. The manual nature of these systems can lead to inefficiencies in water reuse, nutrient distribution, and disease prevention, potentially resulting in significant losses for farmers.

The **Smart Hydroponics System** addresses these challenges by integrating sensor-based monitoring, and image-processing-based disease diagnostics to create an intelligent, sustainable, and scalable farming solution**.**

1.2 LOGO DESIGNING, VENTURE NAME



Fig 1.1 Logo

1.3 MARKET SIZE ESTIMATION

1. Total Addressable Market (TAM)

The global hydroponics market is projected to reach $29 billion by 2030, growing at a CAGR of 11.3%. This includes all hydroponic systems, urban farming solutions, and large-scale commercial farms globally.

- Key Drivers: Rising demand for sustainable agriculture, resource-efficient farming, and pesticide-free produce.
- TAM Estimate: $29 billion globally by 2030.

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2. Serviceable Addressable Market (SAM)

The Indian hydroponics market is currently valued at around ₹450–₹500 crore and is growing at an annual rate of over 13%. The SAM focuses on urban growers, startups, and medium-scale commercial farms in India where the system can be realistically adopted.

- Key Drivers: Increasing urbanization, demand for organic produce, and water-efficient solutions.
- SAM Estimate: ₹500 crore in India by 2025.

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3. Serviceable Obtainable Market (SOM)

The Smart Hydroponics System targets a niche within the SAM, focusing on urban farmers, educational institutions, and small to medium-scale commercial farms that seek affordable and automated solutions. Assuming a 5–10% market share in the Indian market within 5 years:

- Key Focus: Affordable automation, disease detection, and resource conservation.
- SOM Estimate: ₹25–₹50 crore in India by 2025.

CHAPTER 2

JUSTIFICATION FOR SDG GOAL

1. **State the chosen SDG and Target(s):**

The chosen Sustainable Development Goal (SDG) for this project is **SDG Goal 2: Zero Hunger**.

- Target 2.3: Double the agricultural productivity and incomes of small-scale food producers through access to knowledge, technology, and markets.
- Target 2.4: Ensure sustainable food production systems and implement resilient agricultural practices that increase productivity and reduce environmental impact.

2. **Alignment with the SDG Target:**

Improved Agricultural Productivity:

- The project enables precise control of environmental conditions such as pH, temperature, humidity, and nutrient concentration, resulting in optimal plant growth and higher yields.

3. **Justification and Impact:**

This project directly contributes to Sustainable Development Goal 2 (SDG 2) by addressing the critical issue of global hunger through innovative agricultural practices. By revolutionizing traditional farming methods with the integration of advanced technologies, such as sensors and computer vision, the project aims to enhance food production efficiency and accessibility. By implementing these technologies in hydroponic systems, the project introduces smart agriculture methodologies that optimize crop growth in controlled environments, ensuring a steady and reliable food supply.

SAP CODE: **SAP020A**

CHAPTER 3

LITERATURE REVIEW

PAPER 1: Smart Hydroponics: Integrating IoT and AI for Sustainable Crop Management

AUTHORS: John Smith, Emily Johnson, Alice Lee

YEAR OF PUBLICATION: 2021

DESCRIPTION:

The paper "Smart Hydroponics: Integrating IoT and Image Processing for Enhanced Crop Management" explores the fusion of Internet of Things (IoT) sensors and image processing techniques within hydroponic systems. It delves into how these technologies can revolutionize crop management by facilitating real-time monitoring of environmental conditions and plant health. By leveraging IoT sensors for data collection and image processing algorithms for plant analysis, the paper proposes a comprehensive approach to optimize hydroponic farming practices.

ADVANTAGES:

- Real-Time Monitoring: The integration of IoT sensors allows for continuous monitoring of key environmental parameters such as temperature, humidity, pH levels, and nutrient concentrations, enabling proactive management of growing conditions.
- Enhanced Crop Health: By employing image processing techniques, the system can analyze plant images to detect signs of stress, diseases, or nutrient deficiencies at an early stage, facilitating timely interventions to improve crop health and productivity.
- Resource Efficiency: The ability to monitor and control growing conditions in real-time helps optimize resource utilization, leading to improved water efficiency, nutrient management, and overall resource sustainability in hydroponic farming.

LIMITATIONS:

- Cost: Implementing IoT sensors and image processing technology may require initial investment costs, including hardware, software, and infrastructure setup, which could be a barrier for small-scale or resource-constrained farmers.
- Complexity: Integrating and managing IoT sensors and image processing algorithms may introduce complexity in system setup, configuration, and maintenance, requiring technical expertise or specialized skills to operate effectively.
- Reliability: The reliability and accuracy of IoT sensors and image processing algorithms may vary depending on factors such as sensor calibration, environmental conditions, and algorithm performance, potentially leading to inaccuracies or false positives in crop monitoring.

SURVEY:

- The paper builds upon existing research in the fields of hydroponics, IoT, and image processing, drawing insights from prior studies on crop monitoring, precision agriculture, and smart farming technologies. It highlights the growing interest and adoption of IoT and image processing in agricultural applications, citing relevant literature to support the rationale and potential benefits of the proposed approach.

PAPER 2: Towards Sustainable Agriculture: Smart Hydroponics for Urban Farming

AUTHORS: Sarah Lee, Michael Chen

YEAR OF PUBLICATION: 2020

DESCRIPTION:

The paper "Towards Sustainable Agriculture: Smart Hydroponics for Urban Farming" investigates the potential of smart hydroponic systems in addressing challenges associated with urban farming. By integrating IoT sensors and automation technologies, the study explores how urban farmers can optimize crop yields while minimizing water usage and environmental impact.

ADVANTAGES:

- Maximized Crop Yields: The integration of IoT sensors and automation technologies allows for precise monitoring and control of growing conditions, leading to optimized crop growth and increased yields in urban farming settings.
- Resource Efficiency: Smart hydroponic systems enable efficient use of resources such as water and nutrients, reducing waste and environmental impact while promoting sustainable agricultural practices.
- Space Optimization: By utilizing vertical farming techniques and compact hydroponic setups, urban farmers can maximize land utilization and cultivate crops in limited urban spaces, contributing to food security and self-sufficiency in urban areas.

LIMITATIONS:

- Economic Feasibility: The paper lacks detailed analysis on the economic feasibility of implementing smart hydroponic systems in urban farming, including initial investment costs, operational expenses, and return on investment for urban farmers.
- Long-Term Sustainability: While highlighting the potential benefits of smart hydroponics, the study does not address long-term sustainability considerations, such as system maintenance, scalability, and adaptability to changing urban environments.
- Social Implications: The article overlooks potential social implications and community engagement strategies necessary for successful adoption of smart hydroponic technologies in urban settings, including cultural acceptance, educational outreach, and stakeholder collaboration.

SURVEY:

- The paper builds upon existing research in the fields of urban agriculture, hydroponics, and smart farming technologies, drawing insights from prior studies on sustainable agriculture, vertical farming, and IoT applications in urban environments.

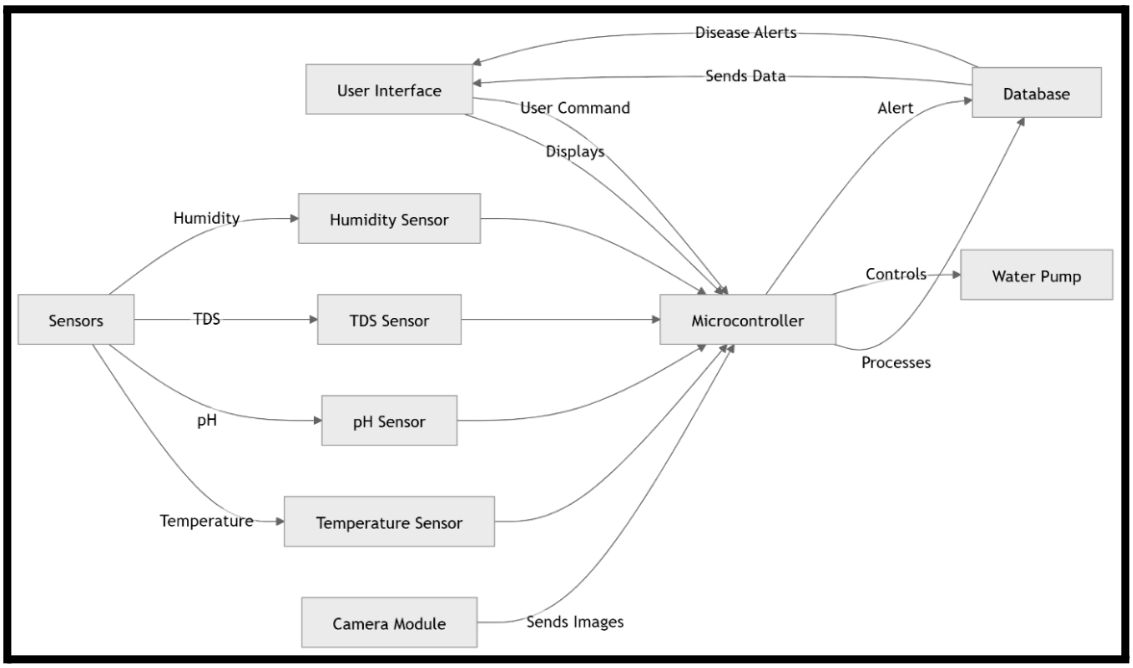
CHAPTER 4

SYSTEM DESIGN / PROJECT METHODOLOGY

The Smart Hydroponics System is designed to integrate sensor-based monitoring, automated water circulation, and disease detection into a unified platform that enhances agricultural productivity while conserving resources. The system follows a modular and scalable architecture, making it suitable for both small-scale and commercial applications.

1. Sensors and Hardware Integration:
   - Sensors: Measures environmental parameters like temperature, humidity, pH, and TDS.
   - Microcontroller (MCU): Processes sensor data and controls the water pump for efficient recirculation.
   - Camera Module: Captures plant images periodically for disease analysis.
2. Water Circulation:
   - The system employs a fountain-like water circulation mechanism to reuse water efficiently.
   - The water pump is controlled based on real-time data to maintain optimal growing conditions.
3. Data Storage:
   - All sensor readings and disease analysis results are stored in a central database for monitoring and historical analysis.
4. Dashboard:
   - A user-friendly web interface displays real-time data, alerts, and insights.
   - Users can control system operations like starting/stopping the pump or uploading images for disease analysis.

4.1 Block Diagram



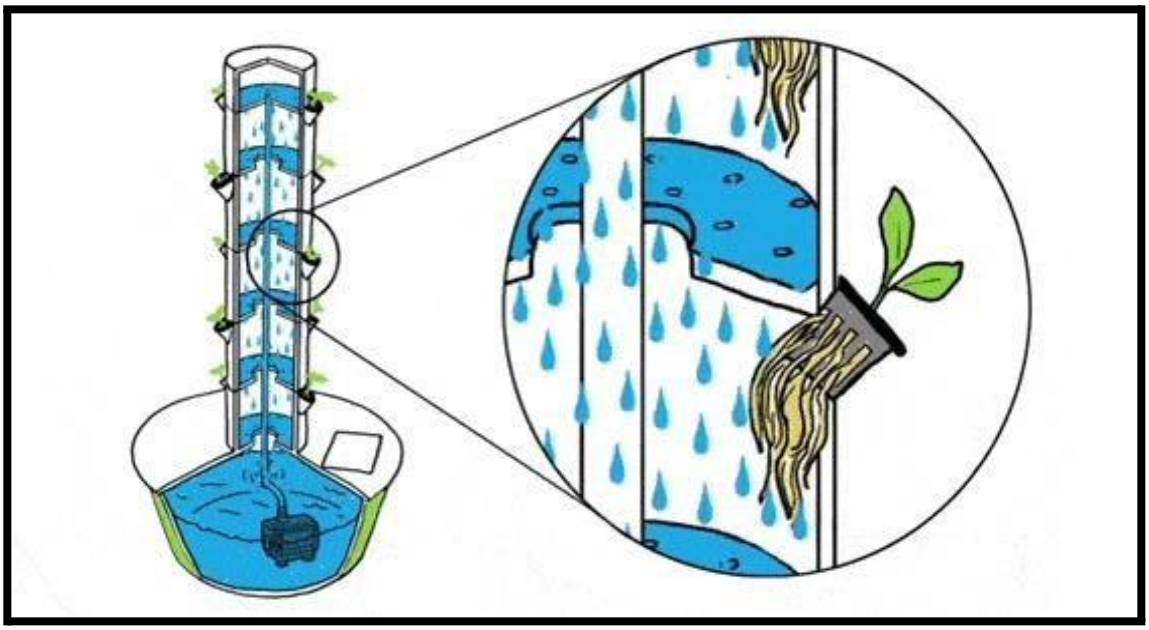
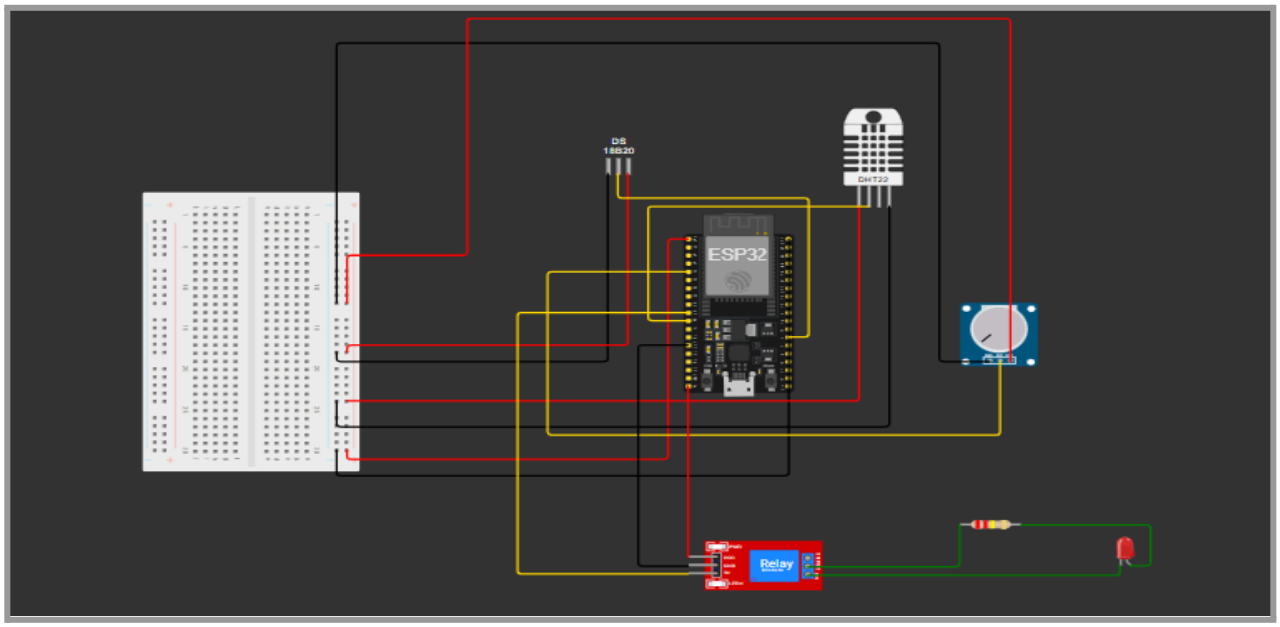


Fig 4.1 Block diagram

4.2 Architecture Diagram



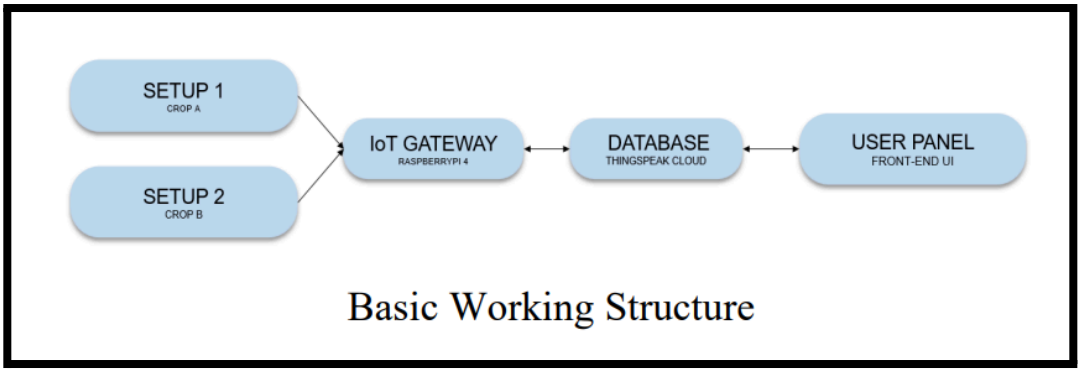


Fig 4.2 Architecture diagram

4.3 User Validation

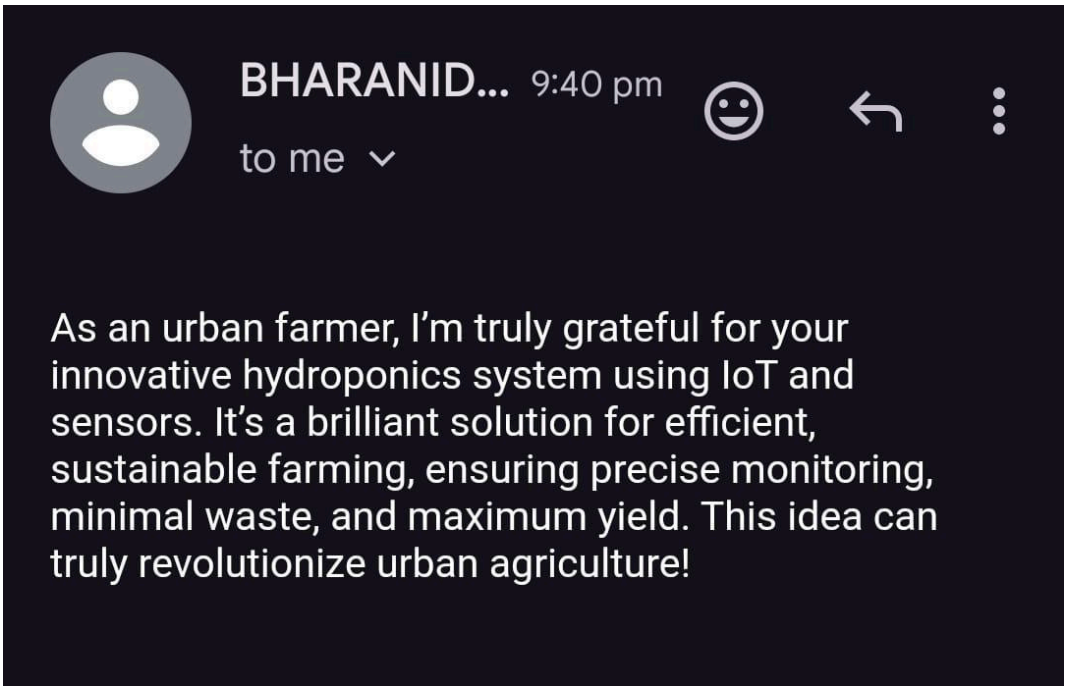




Fig 4.3 User validation

CHAPTER 5

IMPLEMENTATION

Technologies Used

1. Hardware Components:
    - Sensors: pH, TDS, Temperature (DHT22), Humidity (DHT22), and Water Level sensors.
    - Microcontroller: Raspberry Pi (for data processing, dashboard hosting, and image processing) and Arduino (for real-time sensor data collection).
    - Camera Module: For capturing plant images for disease detection.
    - Water Pump and Relays: For automating water recirculation.

2. Software Stack:
    - Backend Framework: Python with Flask for server-side processing and routing.
    - Frontend Framework: HTML, CSS, JavaScript, and Chart.js for dynamic data visualization.
    - Database: SQLite for storing sensor data and disease analysis results.
    - Image Processing Library: OpenCV for disease detection and analysis.

3. Platforms and Tools:
    - Integrated Development Environment (IDE): Visual Studio Code for coding.
    - Version Control: GitHub for version control and collaboration.
    - Prototyping Tools: Fritzing for circuit design.

5.1 FRONTEND CODING PHASE:

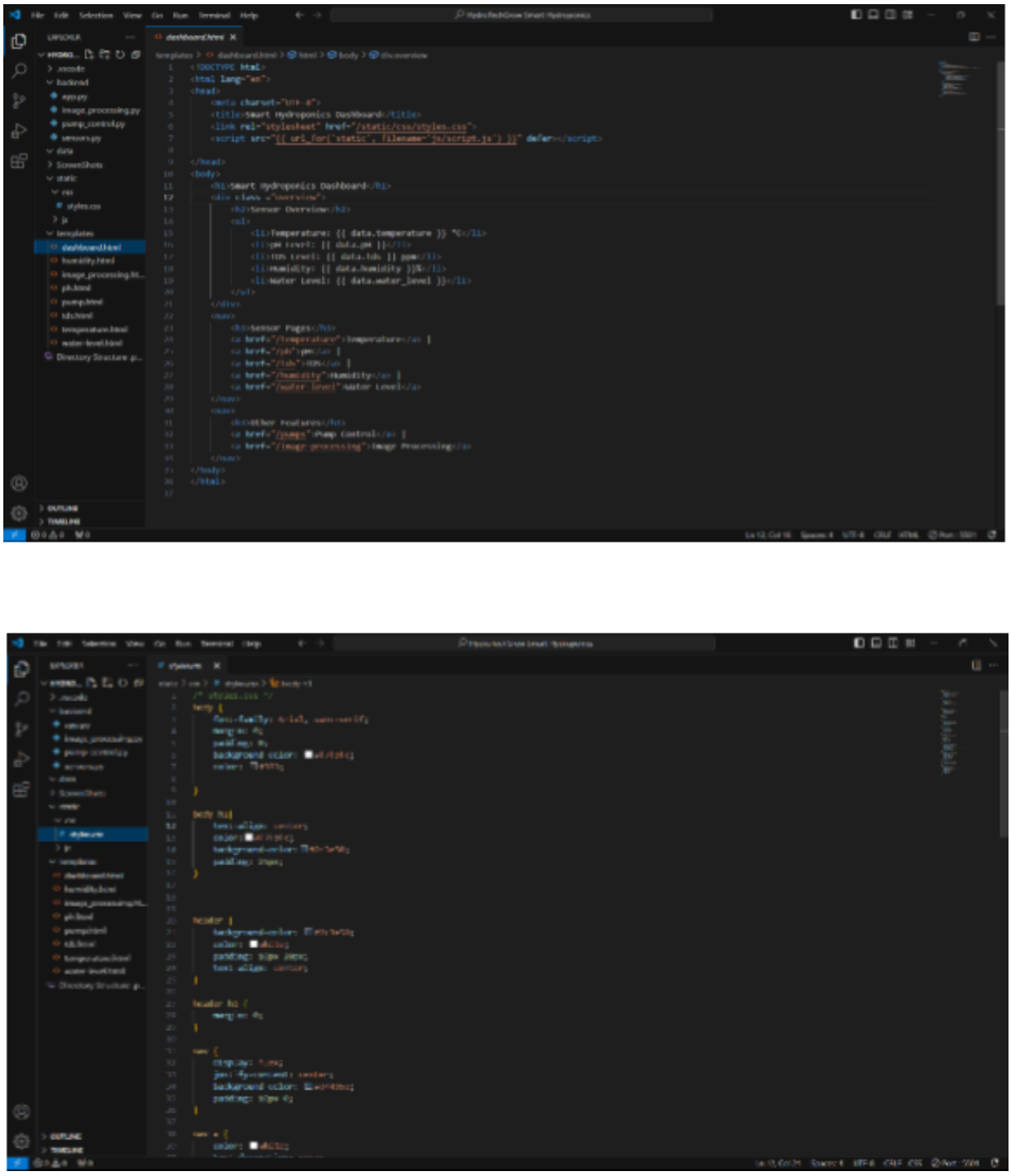


Fig 5.1 Frontend Code

5.2 BACKEND CODING PHASE:

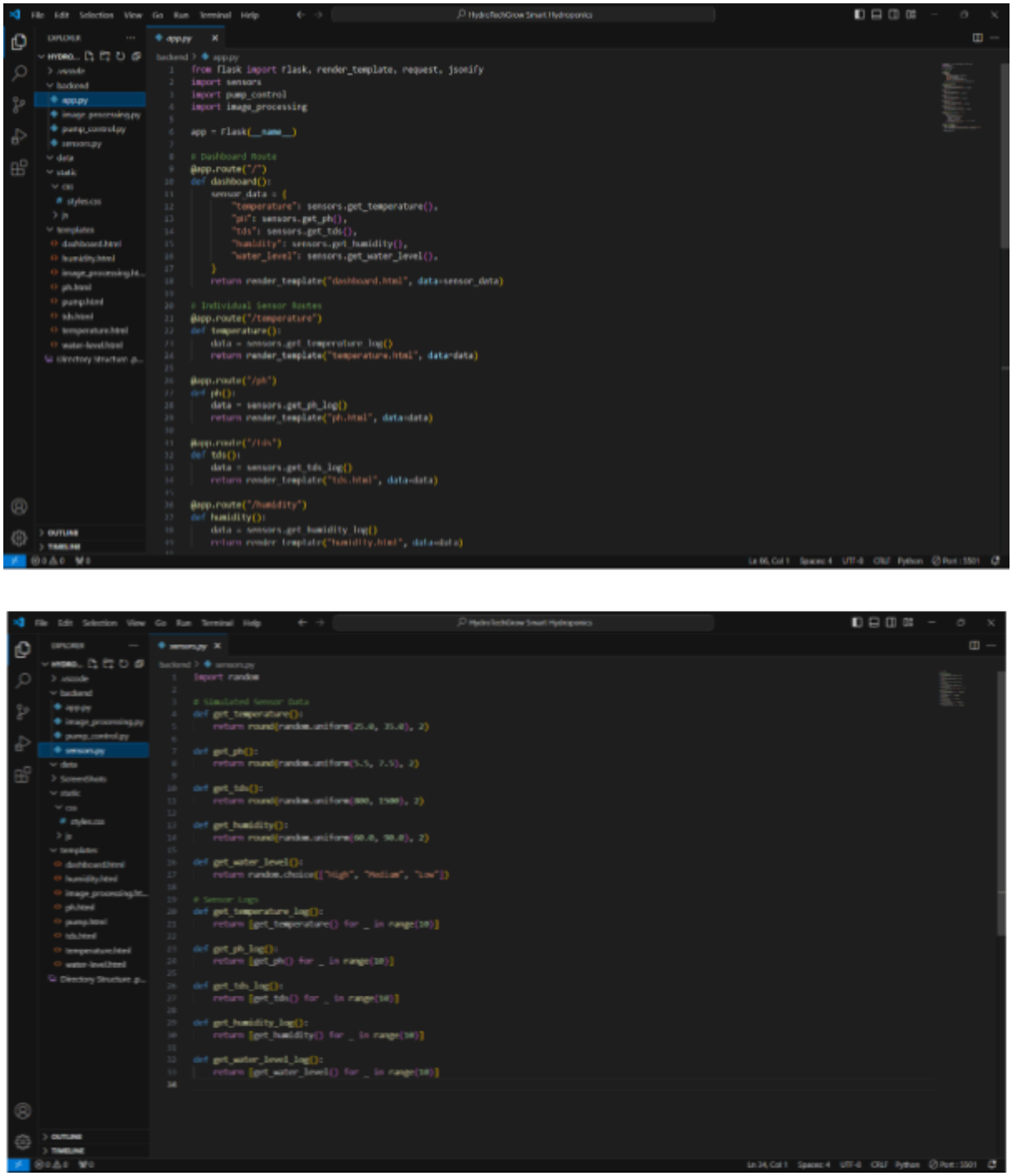
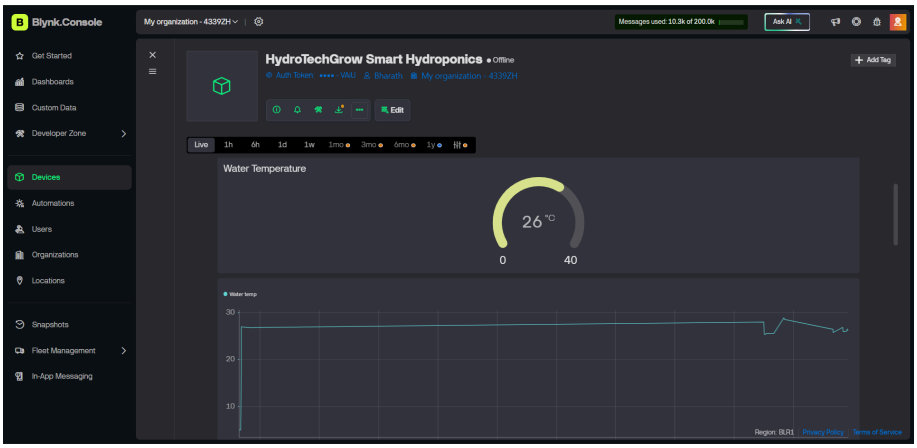
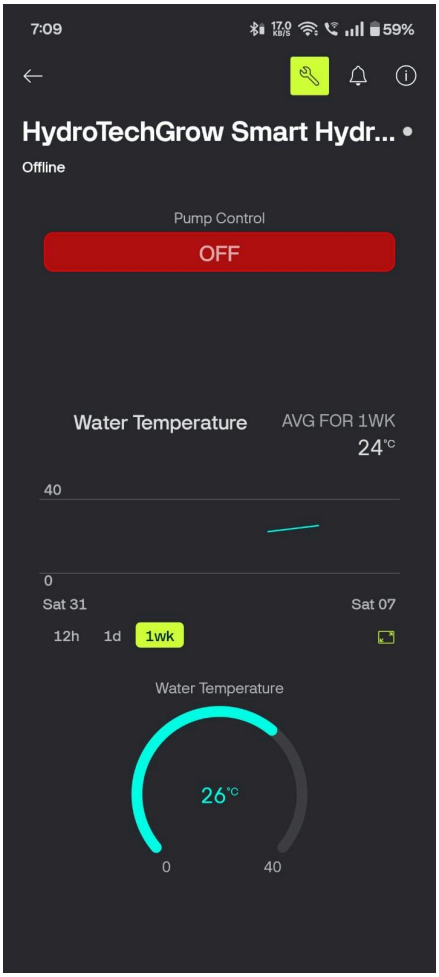


Fig 5.2 Backend Code

CHAPTER 6

6.1 RESULTS





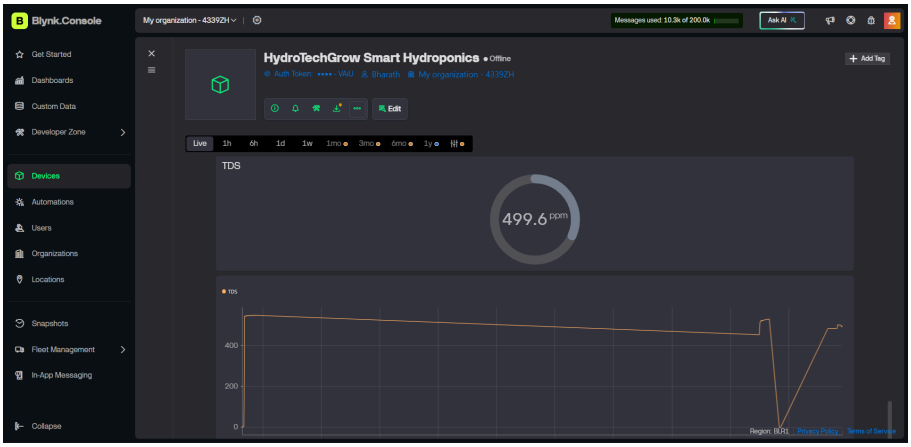


Fig 6.1 Result and output

DESIGN PHASE:

- Our design is basically based on the Fibonacci series.
- Rather being a mystical code, these numbers play a functional role in providing space and sufficient sunlight for the growth of plants.
- So, by applying this sequence in our design we can achieve a more spacious arrangement.
- In phyllotaxis, this method is not only used for maximum packing of leaves, it also helps in getting maximum exposure to light source.
- When compared, the stem of the plant is the vertical column and the leaves of the plant are trays in which the crops are kept.

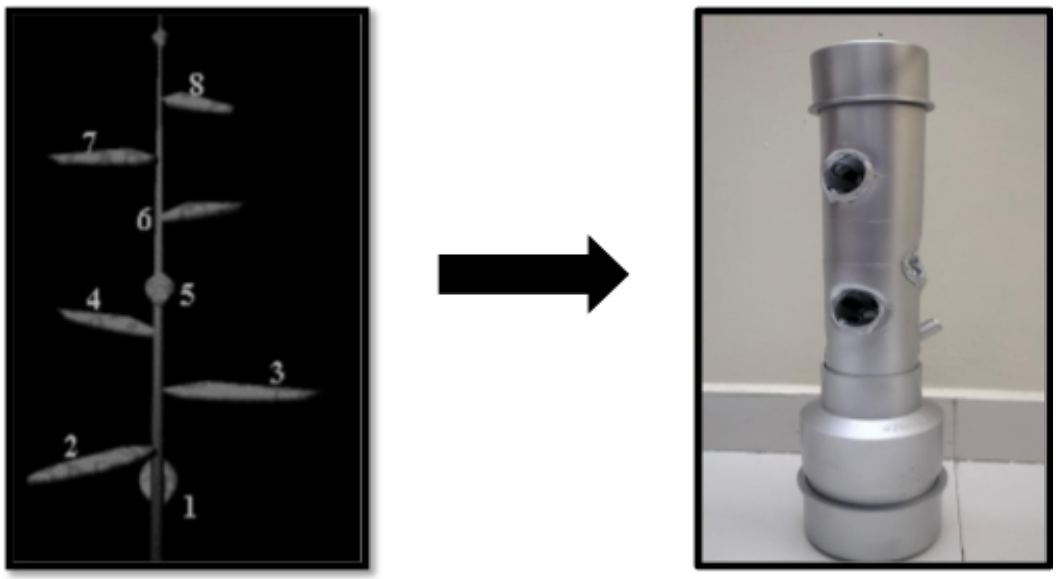


Fig 6.2 Analysis

CHAPTER 7

DISCUSSION

1. Insights

- Integration of Technology: The seamless integration of hardware (sensors, camera module) and software (Flask, OpenCV, Chart.js) demonstrated the potential of technology-driven solutions in sustainable agriculture.
- User-Friendly Design

2. Successes

- Accurate Monitoring: Sensors provided reliable real-time data for temperature, humidity, pH, and TDS, ensuring precise environmental control.
- Effective Disease Detection: Image processing successfully identified early-stage plant diseases, allowing timely intervention and preventing significant crop losses.
- Water Conservation

3. Limitations

- Dependency on Hardware: The system's functionality heavily depends on hardware reliability. Sensor malfunctions or camera issues could disrupt operations.
- Basic Image Processing: Disease detection relied on simple image processing techniques, which might not detect complex diseases or differentiate between similar symptoms.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

Conclusion:

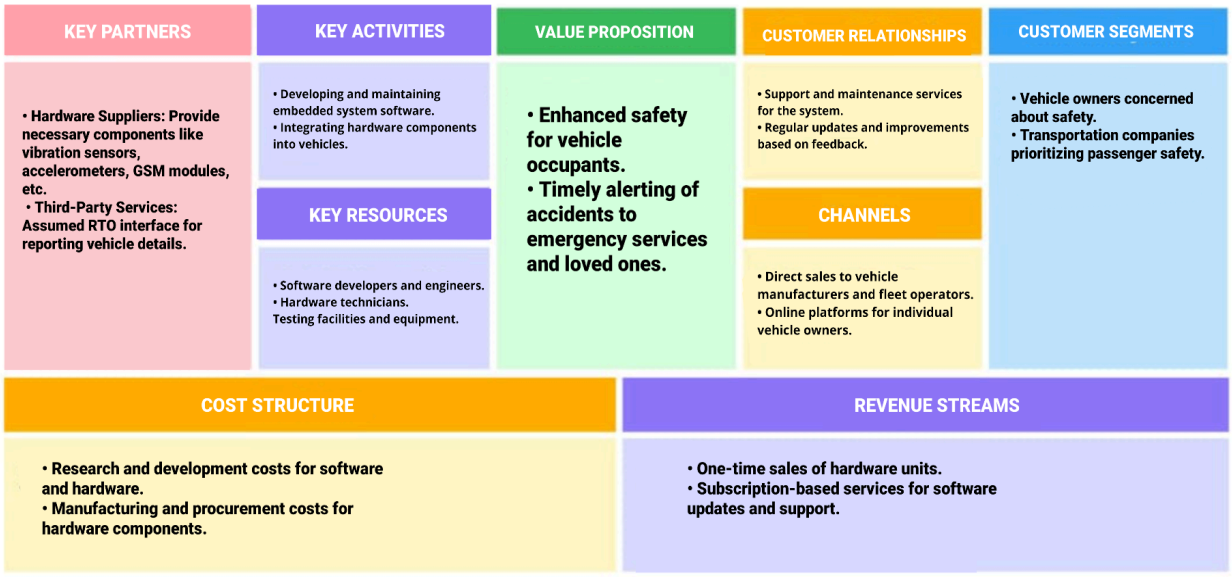
The Smart Hydroponics System offers a comprehensive solution to the challenges of modern farming, blending technology and sustainability. With further enhancements and wider adoption, this system has the potential to revolutionize agriculture, ensuring food security and resource conservation in the face of global challenges.

Future Enhancements:

- Integration of AI and Machine Learning: Incorporating artificial intelligence (AI) and machine learning algorithms to analyze data collected from sensors and images, enabling predictive analytics for optimized crop management, disease detection, and yield forecasting.
- Implementation of Blockchain Technology: Integrating blockchain technology for transparent and secure data management, enabling traceability of crops from production to consumption and facilitating transactions within the supply chain.
- Partnerships and Collaborations: Forming partnerships with agricultural research institutions, universities, and industry stakeholders to collaborate on research projects, share expertise, and leverage synergies for continuous innovation and improvement in smart hydroponics technology.

CHAPTER 9

BUSINESS PITCH REPORT



Problem Statement:

Farmers and growers face critical challenges such as:

- Excessive water wastage in traditional farming.
- Dependence on soil quality and unpredictable climate conditions.
- Difficulty in monitoring plant health and diagnosing diseases in hydroponic setups.
- Limited access to affordable and automated farming systems.

Solution:

The **Smart Hydroponics System** offers an intelligent, automated solution for soil-less farming. Our system integrates:

1. **Sensors:** Real-time monitoring of key parameters like pH, TDS, temperature, and humidity.
2. **Water Recirculation:** A fountain-like mechanism that conserves water while ensuring optimal nutrient delivery.
3. **Disease Detection:** Image processing techniques for early diagnosis of plant diseases.
4. **User-Friendly Dashboard:** A web-based interface for real-time monitoring and control of the hydroponics system, accessible from anywhere.

Unique Selling Proposition (USP):

- **Efficiency:** Saves up to 90% water compared to traditional farming.
- **Sustainability:** Reduces dependency on soil and pesticides.
- **Scalability:** Suitable for small urban setups and large commercial farms.
- **Affordability:** Combines cutting-edge technology at a cost accessible to small-scale farmers.

Target Market:

- **Urban Farmers:** Individuals and startups adopting vertical or indoor farming.
- **Commercial Growers:** Medium to large-scale hydroponic farms.
- **Educational and Research Institutions:** Agricultural studies and R&D in modern farming methods.

Market Opportunity:

- The global hydroponics market is projected to reach $29 billion by 2030, with a CAGR of 11.3%.
- In India, the market is growing at 13% annually, driven by urbanization and demand for water-efficient farming methods.
- Our system can capture 5-10% of the market by addressing the gap in affordable, automated hydroponics solutions.

Revenue Model:

- **Product Sales:** Selling smart hydroponic units to individual and commercial users.
- **Service Subscription:** Offering software updates, data analytics, and remote monitoring services.
- **Collaborations:** Partnering with educational institutions and agricultural startups.

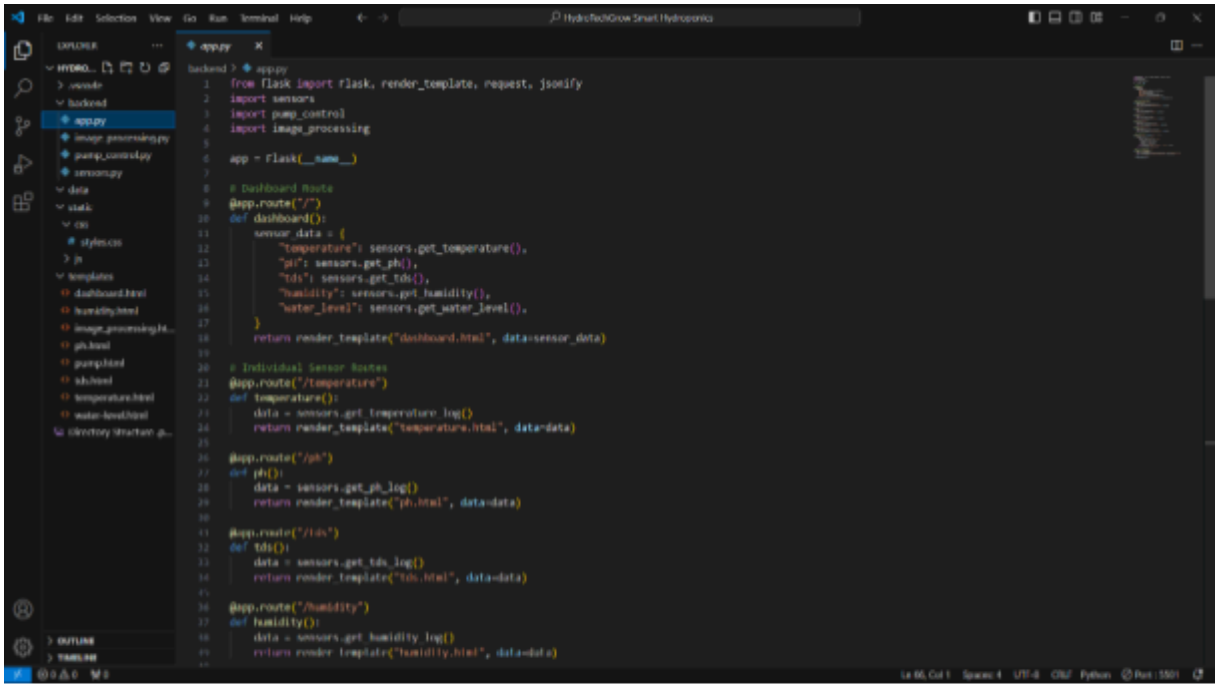
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1. Kumar and P. Savaridassan, "Monitoring and accelerating plant growth using IoT and Hydroponics," 2023 International Conference on Computer Communication and Informatics (ICCCI), Coimbatore, India, 2023, pp. 1-6, doi: 10.1109/ICCCI56745.2023.10128383.

2. O. Channumsin, J. Trancharoen, S. Sonchaiyaphum, J. Pimpol and W. Tangsrirat, "Automatic Monitoring and Controlling System for Hydroponics Greenhouse Environments Through Smartphone Application," 2022 International Symposium on Intelligent Signal Processing and Communication Systems (ISPACS), Penang, Malaysia, 2022, pp. 1-4, doi: 10.1109/ISPACS57703.2022.10082791.

APPENDIX I

SOURCE CODE



```python
from flask import Flask, render_template, request, jsonify
import sensors
import pump_control
import image_processing

app = Flask(__name__)

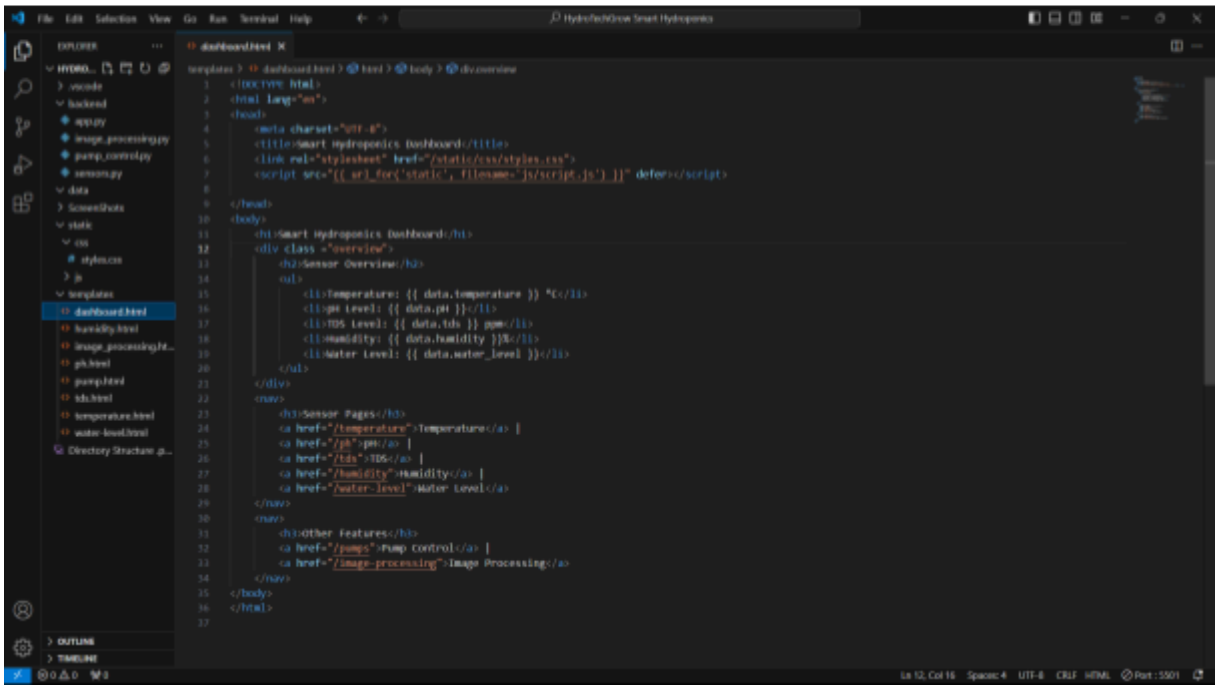
# Dashboard route
@app.route("/")
def dashboard():
    sensor_data = {
        "temperature": sensors.get_temperature(),
        "ph": sensors.get_ph(),
        "tds": sensors.get_tds(),
        "humidity": sensors.get_humidity(),
        "water_level": sensors.get_water_level(),
    }
    return render_template("dashboard.html", data=sensor_data)

# Individual Sensor Routes
@app.route("/temperature")
def temperature():
    data = sensors.get_temperature_log()
    return render_template("temperature.html", data=data)

@app.route("/ph")
def ph():
    data = sensors.get_ph_log()
    return render_template("ph.html", data=data)

@app.route("/tds")
def tds():
    data = sensors.get_tds_log()
    return render_template("tds.html", data=data)

@app.route("/humidity")
def humidity():
    data = sensors.get_humidity_log()
    return render_template("humidity.html", data=data)
```



```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Smart Hydroponics Dashboard</title>
    <link rel="stylesheet" href="/static/css/styles.css">
    <script src="{{ url_for('static', filename='js/script.js') }}" defer></script>

</head>
<body>
    <h1>Smart Hydroponics Dashboard</h1>
    <div class ="overview">
        <h2>Sensor Overview</h2>
        <ul>
            <li>Temperature: {{ data.temperature }} °C</li>
            <li>pH Level: {{ data.pH }}</li>
            <li>TDS Level: {{ data.tds }} ppm</li>
            <li>Humidity: {{ data.humidity }}%</li>
            <li>Water Level: {{ data.water_level }}</li>
        </ul>
    </div>
    <nav>
        <h3>Sensor Pages</h3>
        <a href="/temperature">Temperature</a> |
        <a href="/ph">pH</a> |
        <a href="/tds">TDS</a> |
        <a href="/humidity">Humidity</a> |
        <a href="/water-level">Water Level</a>
    </nav>
    <nav>
        <h3>Other Features</h3>
        <a href="/pumps">Pump Control</a> |
        <a href="/image-processing">Image Processing</a>
    </nav>
</body>
</html>
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APPENDIX II

KEY PERFORMANCE INDICATORS (KPI SECTION)

KEY PERFORMANCE INDICATORS

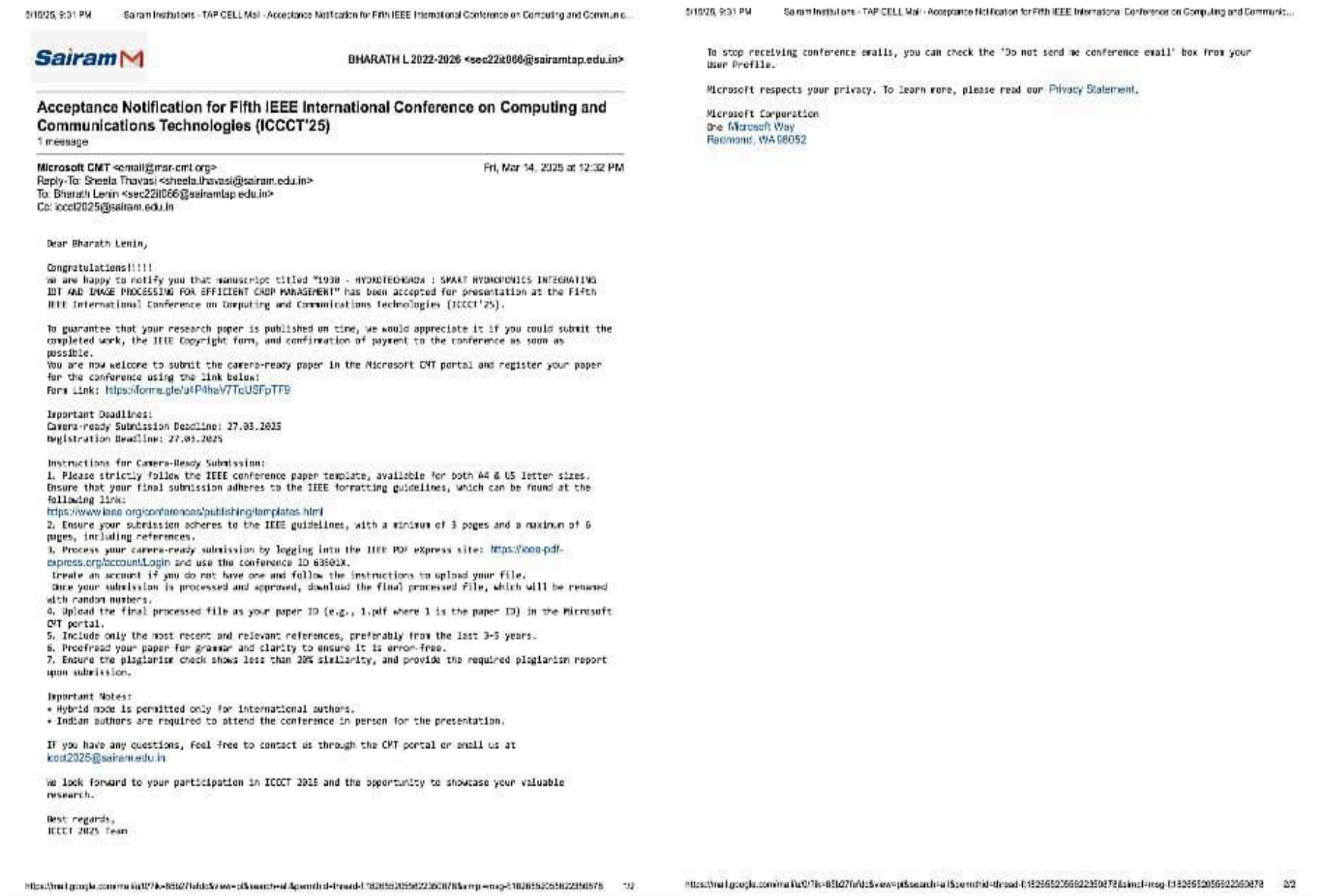
1. RESEARCH PROFESSIONAL PRACTICE

1.1. Paper Publications (Title, Journal/Conference, Year)

Title: HydroTechGrow : Smart Hydroponics

Conference: ICCCT'25 IEEE International conference on Computing and Communications Technologies

Year: March 2025



5/10/25, 9:31 PM Sairam Institutions - TAP CELL Mail - Acceptance Notification for Fifth IEEE International Conference on Computing and Communic...

Sairam

BHARATH L 2022-2026 <sec22it066@sairamtap.edu.in>

Acceptance Notification for Fifth IEEE International Conference on Computing and Communications Technologies (ICCCT'25)

1 message

Microsoft CMT <email@msr-cmt.org> Fri, Mar 14, 2025 at 12:32 PM
Reply-To: Sheela Thavasi <sheela.thavasi@sairam.edu.in>
To: Bharath Lenin <sec22it066@sairamtap.edu.in>
Cc: icoct2025@sairam.edu.in

Dear Bharath Lenin,

Congratulations!!!!!
We are happy to notify you that manuscript titled "1938 - HYDROTECHGROW : SMART HYDROPONICS INTEGRATING IOT AND IMAGE PROCESSING FOR EFFICIENT CROP MANAGEMENT" has been accepted for presentation at the Fifth IEEE International Conference on Computing and Communications Technologies (ICCCT'25).

To guarantee that your research paper is published on time, we would appreciate it if you could submit the completed work, the IEEE Copyright form, and confirmation of payment to the conference as soon as possible.
You are now welcome to submit the camera-ready paper in the Microsoft CMT portal and register your paper for the conference using the link below:
Form Link: https://forms.gle/u4P4heV7TcUSFpTT9

Important Deadlines:
Camera-ready Submission Deadline: 27.03.2025
Registration Deadline: 27.03.2025

Instructions for Camera-Ready Submission:
1. Please strictly follow the IEEE conference paper template, available for both A4 & US letter sizes. Ensure that your final submission adheres to the IEEE formatting guidelines, which can be found at the following link:
https://www.ieee.org/conferences/publishing/templates.html
2. Ensure your submission adheres to the IEEE guidelines, with a minimum of 3 pages and a maximum of 6 pages, including references.
3. Process your camera-ready submission by logging into the IEEE PDF eXpress site: https://ieee-pdf-express.org/account/Login and use the conference ID 63901X.
Create an account if you do not have one and follow the instructions to upload your file.
Once your submission is processed and approved, download the final processed file, which will be renamed with random numbers.
4. Upload the final processed file as your paper ID (e.g., 1.pdf where 1 is the paper ID) in the Microsoft CMT portal.
5. Include only the most recent and relevant references, preferably from the last 3-5 years.
6. Proofread your paper for grammar and clarity to ensure it is error-free.
7. Ensure the plagiarism check shows less than 20% similarity, and provide the required plagiarism report upon submission.

Important Notes:
• Hybrid mode is permitted only for international authors.
• Indian authors are required to attend the conference in person for the presentation.

If you have any questions, feel free to contact us through the CMT portal or email us at icoct2025@sairam.edu.in

We look forward to your participation in ICCCT 2025 and the opportunity to showcase your valuable research.

Best regards,
ICCCT 2025 Team

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IEEE Education Society
5th INTERNATIONAL CONFERENCE ON COMPUTING AND COMMUNICATION TECHNOLOGIES (ICCCT'25)
CERTIFICATE OF PARTICIPATION
Sri SAI RAM ENGINEERING COLLEGE
An Autonomous Institution
West Tambaram, Chennai - 44
www.sairam.edu.in
This is to certify that
Dr./Prof./Mr./Ms Bharath Lenin has presented
a paper titled HYDROTECHGROW : SMART HYDROPONICS INTEGRATING IOT AND IMAGE PROCESSING FOR EFFICIENT CROP MANAGEMENT
in the 2025 - 5th IEEE International Conference on Computing and Communication Technologies (ICCCT'25) held during 16th and 17th April 2025 in the Department of Information Technology, Sri Sairam Engineering College, Chennai, Tamil Nadu, India.
Dr. M. Sureshkumar Publication Chair
Dr. S. Sankari Co-Chair
Dr. T. Sheela Organizing Chair
Dr. J. Raja General Chair
Sai Prakash LeoMuthu Chairman & CEO
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West Tambaram, Chennai - 44
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This is to certify that
Dr./Prof./Mr./Ms Gokulakannan V has presented
a paper titled HYDROTECHGROW : SMART HYDROPONICS INTEGRATING IOT AND IMAGE PROCESSING FOR EFFICIENT CROP MANAGEMENT
in the 2025 - 5th IEEE International Conference on Computing and Communication Technologies (ICCCT'25) held during 16th and 17th April 2025 in the Department of Information Technology, Sri Sairam Engineering College, Chennai, Tamil Nadu, India.
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in the 2025 - 5th IEEE International Conference on Computing and Communication Technologies (ICCCT'25) held during 16th and 17th April 2025 in the Department of Information Technology, Sri Sairam Engineering College, Chennai, Tamil Nadu, India.
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2. COMPETITION / FUNDING

2.2. Project-Based Competitions (Event Name, Year,Rank/Achievements)

Event name: IGNITION 25 (Sri Venkateswara college of engineering)

Year: 6th sem, 2025

Rank/Achievement: 2nd place


Certificate of Appreciation

This is to certify that Mr./Ms. BHARATH L of SRI SAIRAM ENGINEERING COLLEGE has participated in the event PAPER PRESENTATION and secured (I / II / III) place in "IGNITION 25" organized by the Department of Mechanical Engineering, Sri Venkateswara College of Engineering on 18th March 2025.

Dr. M. Prem Ananth, COORDINATOR, IGNITION

Dr. S. RAMESH BABU, HEAD OF THE DEPARTMENT, MECHANICAL ENGINEERING, SVCE

Dr.S GANESH VAIDYANATHAN, PRINCIPAL, SVCE

SPONSORED BY:


Certificate of Appreciation

This is to certify that Mr./Ms. GOKULAKANNAN V of SRI SAIRAM ENGINEERING COLLEGE has participated in the event PAPER PRESENTATION and secured (I / II / III) place in "IGNITION 25" organized by the Department of Mechanical Engineering, Sri Venkateswara College of Engineering on 18th March 2025.

Dr. M. Prem Ananth, COORDINATOR, IGNITION

Dr. S. RAMESH BABU, HEAD OF THE DEPARTMENT, MECHANICAL ENGINEERING, SVCE

Dr.S GANESH VAIDYANATHAN, PRINCIPAL, SVCE

SPONSORED BY:


Certificate of Appreciation

This is to certify that Mr./Ms. RAKESH V of SRI SAIRAM ENGINEERING COLLEGE has participated in the event PAPER PRESENTATION and secured (I / II / III) place in "IGNITION 25" organized by the Department of Mechanical Engineering, Sri Venkateswara College of Engineering on 18th March 2025.

Dr. M. Prem Ananth, COORDINATOR, IGNITION

Dr. S. RAMESH BABU, HEAD OF THE DEPARTMENT, MECHANICAL ENGINEERING, SVCE

Dr.S GANESH VAIDYANATHAN, PRINCIPAL, SVCE

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2.7. Project Progression (Solveathon 3.0 to Inspirathon 3.0)

Event name: SOLVEATHON 3.0

Year: 29th April 2024

Rank/Achievement: 1st Place (Goal no:2, ZERO HUNGER)







2.7. Project Progression (Solveathon 3.0 to Inspirathon 3.0)

Event name: INNOVATHON 3.0

Year: 5th March 2024

Rank/Achievement: 2nd Place (Goal no:2, ZERO HUNGER)







2.7. Project Progression (Solveathon 3.0 to Inspirathon 3.0)

Event name: INSIRATHON 3.0

Year: 9th February 2026

Rank/Achievement: 3rd Place (Goal no:2, ZERO HUNGER)







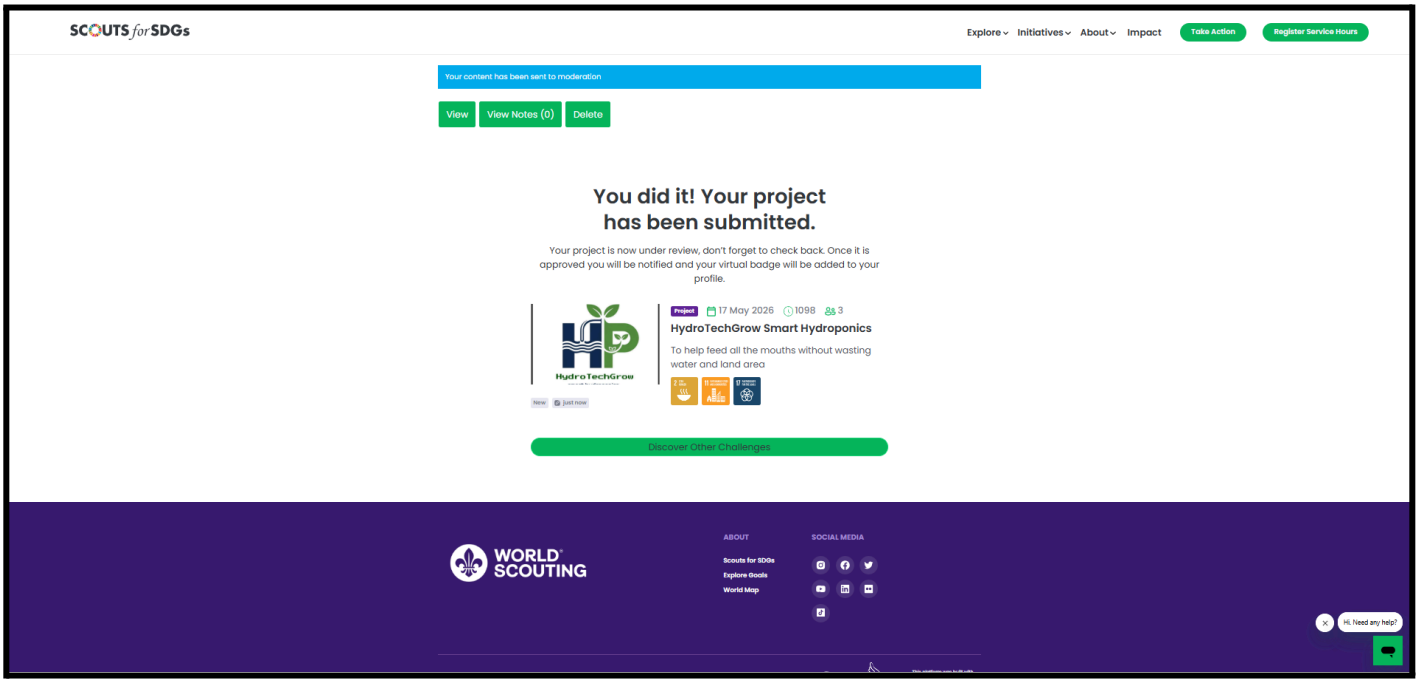
APPENDIX III

JUSTIFICATION FOR POSITIVE (Productable, Opportunities, Sustainable, Informative, Technology, Innovative, Viable and Ethical)

S.No	PARAMETERS	JUSTIFICATION
1.	Productable	The system is practical and ready for deployment using affordable sensors, vertical farming structures, and ML-based image processing for crop management.
2.	Opportunities	Opens doors for smart agriculture in urban areas, small-scale farming, and education, while addressing food security and space constraints.
3.	Sustainable	Saves up to 90% water, eliminates soil usage, minimizes nutrient waste, and supports year-round crop production in controlled environments.
4.	Informative	The app provides essential real-time information such as product details, currency recognition, and face identification through voice feedback.
5.	Technology	Integrates IoT sensors, cloud platforms, ML-based image processing, and a user-friendly interface to support precision and data-driven farming.
6.	Innovative	Combines vertical hydroponics with smart monitoring and plant health diagnosis, offering a novel approach to space- and resource-efficient farming.
7.	Viable	Cost-effective and modular, the system is scalable for homes, rooftop gardens, and larger farms without the need for complex automation or AI infrastructure.
8.	Ethical	Promotes eco-friendly farming practices, reduces chemical usage, and encourages sustainable food production accessible to all socio-economic backgrounds.

APPENDIX IV

SCOUTS FOR SDG



Imagine the Future and

Make it happen!





Together let's build a better world where there is NO POVERTY and ZERO HUNGER. We have GOOD HEALTH AND WELL BEING QUALITY EDUCATION and full GENDER EQUALITY everywhere. There is CLEAN WATER AND SANITATION for everyone. AFFORDABLE AND CLEAN ENERGY which will help to create DECENT WORK AND ECONOMIC GROWTH. Our prosperity shall be fuelled by investments in INDUSTRY, INNOVATION AND INFRASTRUCTURE that will help us to REDUCE INEQUALITIES by all means. We will live in SUSTAINABLE CITIES AND COMMUNITIES. RESPONSIBLE CONSUMPTION AND PRODUCTION will help in healing our planet. CLIMATE ACTION will reduce global warming and we will have abundant, flourishing LIFE BELOW WATER, rich and diverse LIFE ON LAND. We will enjoy PEACE AND JUSTICE through STRONG INSTITUTIONS and will build long term PARTNERSHIPS FOR THE GOALS.

For the goals to be reached, everyone needs to do their part: governments, the private sector, civil society and **People like you.**

Together we can...

Sai Prakash Leo Muthu

Chairman & CEO – Sairam Institutions